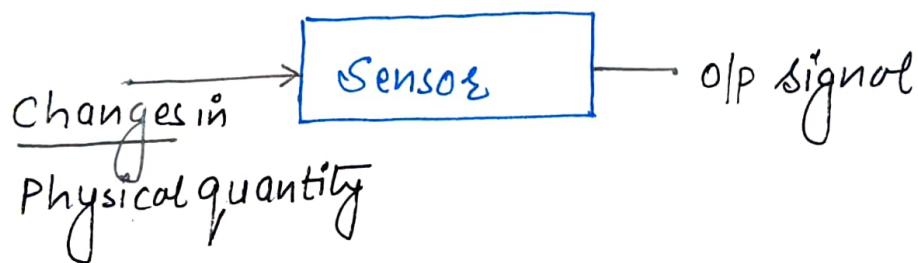


Sensor:- A sensor is a input device which provides an o/p signal of measurable quantity with respect to changes in physical quantity.



eg: ① Thermometer - which senses the changes in body temperature & produces a readable output which gives us information; how much fever a person has.

② Barometer - which senses the changes in atmospheric pressure & correspondingly gives us readable o/p.

Types of Sensors:-

- ① To measure Temperature → We use RTDs, Thermistor & Thermocouple
- ② To measure strain on an object → strain Gauges
- ③ To measure displacement → LVDT sensors
- ④ To measure Vibration & shock → Accelerometers
- ⑤ To detect light → optical sensors

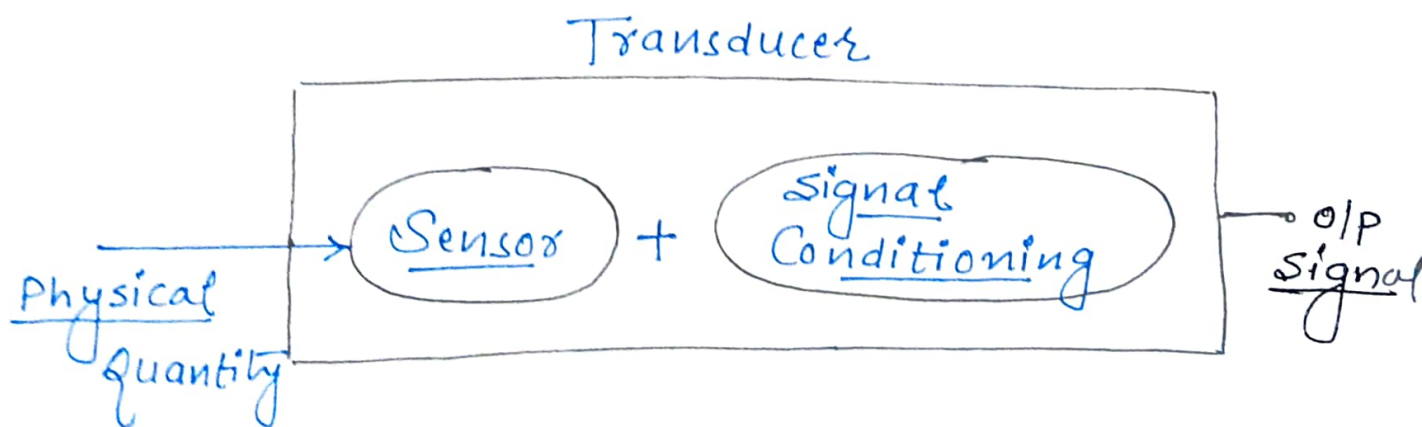
(2)

Active sensor: - Active sensors require an external excitation signal to produce readable o/p.

Passive sensor: - It doesn't require an external excitation signal.

Transducer: - Transducer is a device which transforms one form of energy to another form.

⇒ The process of Conversion of energy from one form to another form is called Transduction.



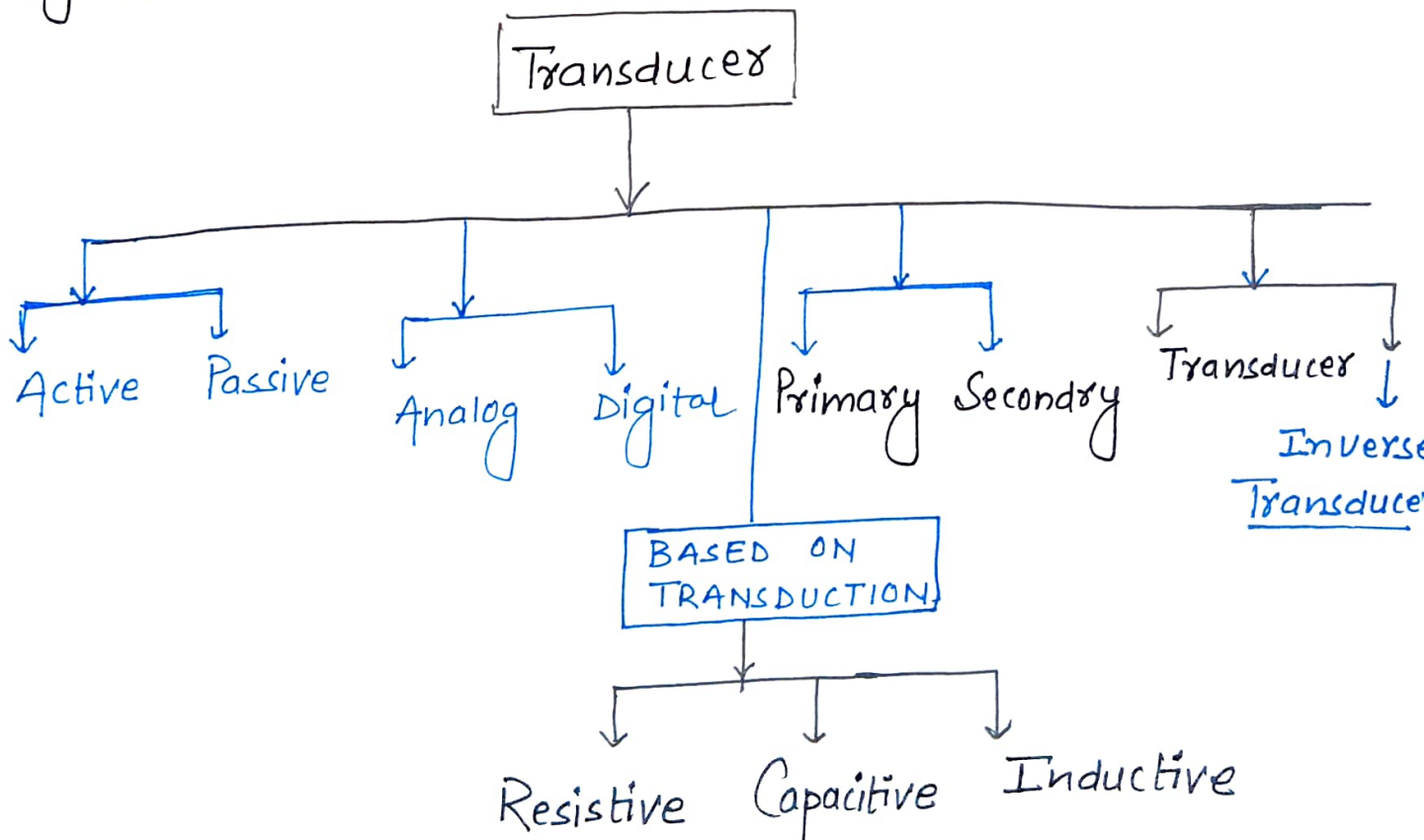
⇒ Sensor senses the physical quantity & signal conditioner strengthens that physical quantity for further processing.

Applications: ① It is used for detecting the movement of muscles which is called Acceleromyograph. ③

② Transducer is used in speaker which converts the electrical signal into acoustic sound.

③ solar Emergency Light — which receives the light energy during day time and gives us electrical o/p during night.

Types of Transducers:-



# Difference between Sensor & Transducer

④

## Sensor

- ① Sensor senses the changes in physical quantity & provides a readable o/p.
- ② Component: Sensor itself
- ③ Examples:- Accelerometer, Barometer etc.
- ④ A sensor may not be a transducer.
- ⑤ Sensor is less complicated.

## Transducer

① Transducer is a device which transforms one form of energy into another form.

② Component: Sensor & signal conditioning.

③ Examples: Thermistor, Thermocouple etc.

④ A transducer will always contain a sensor.

⑤ Transducer is comparatively more complicated.